

Wen-Kai "SansWord" Huang

Senior Software Engineer · Data Infrastructure · Distributed Systems · AI-Assisted Engineering

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"A data-platform builder applying senior engineering discipline
— measurement, cost, judgment — to AI-assisted software."

PROFESSIONAL SUMMARY

Senior Software Engineer with **9+ years** designing and operating large-scale distributed systems and global data infrastructure on **AWS and GCP**, serving **~900M monthly active users** across US, APAC, and Europe. Specialized in high-throughput, fault-tolerant pipelines (>20K msgs/sec), scalable data platforms, and cloud-native migrations. Proven technical leader recognized for cross-functional collaboration and large-scale impact. Now **building and measuring** agentic development and harness engineering through hands-on projects and public writing.

PROFESSIONAL EXPERIENCE

Yahoo!

Sr. Software Dev Engineer · Sunnyvale, CA · Feb 2019 – Jun 2025

- Led end-to-end delivery of Yahoo's data infrastructure evolution to GCP, leading a 5+ engineer team to build a GCP-native ingestion platform using **Cloud Composer, Airflow, and Dataproc** — improving data freshness from 3+ hours to **under 60 minutes** across US, APAC, and Europe.
- Supported a **zero-downtime AWS migration** with Kubernetes-based auto-scaling, maintaining multi-region lambda architecture serving recommendation, ads, and downstream systems across News, Sports, and Finance.
- Built aggregation and caching layers that **reduced data scanned from 725 GB to 121 KB**, cutting query costs by **70%** on Yahoo's BigQuery infrastructure.
- Drove on-call reliability initiative, reducing open incident tickets **from 1,000+ to zero**; practices adopted by the broader on-call team.
- Co-invented a patented salient entity algorithm ([US11803605B2](#)) integrating ranking models into production recommendation systems.

Yahoo! E-Commerce Platform

Sr. Engineer · Taipei, Taiwan · Jun 2016 – Feb 2019

- Drove early AWS adoption across Yahoo engineering, building **blue/green deployment** and auto-scaling patterns via ECS, later adopted by global teams.
- Contributed to migrating **80 TB** to S3 with zero downtime using dual-write/dual-read patterns to ensure safe cutover.
- Scaled **Pigeon** (Apache Pulsar message bus) to **20K msgs/sec**; improved MTBF from 3.5 days to **90 days (~26x)**.
- Core developer of **Cupid**, a multi-tenant discount service with a JSON-based rule engine — driving **10K/day coupon redemption growth**.

PATENT

Determining Salient Entities and Generating Salient Entity Tags Based Upon Articles

[US11803605B2](#)

Co-invented methods to identify salient entities at scale for production recommendation systems at Yahoo.

EDUCATION

M.S., Computer Science

Programming Languages and Software Methodology
National Chengchi University, Taipei, Taiwan

B.S., Mathematics

National Chengchi University, Taipei, Taiwan

TECHNICAL SKILLS

Languages	Python, Java, SQL, Shell, Groovy
Cloud – GCP	Dataproc, Cloud Composer, Dataflow, Airflow, BigQuery
Cloud – AWS	S3, ECS, ELB, ElastiCache
Infrastructure	Terraform, Kubernetes, Docker, Redis
Stream Processing	Apache Storm, Apache Pulsar, Kafka
AI Practice	Agentic Programming, Harness Engineering

WRITING & NOTES

[sans_blog — Essays & Learning Notes](#) ↗

Field notes on building software with AI — the skeptic-to-shipping story, system-design study guides, and context-engineering deep-dives.

AI ENGINEERING IN PUBLIC

The Document Tree

Context engineering — a lazy-loaded doc tree that keeps an AI agent's context in sync with the code. Productized as a reusable template, [sans_doc_tree](#).

sans_schema

Semantic query gateway — LLM in the request path behind a validated-AST safety boundary; clients query in their own field names + plain English. FastAPI/Postgres, [live demo](#).

guasi (我是)

Live full-stack identity-verification product — Next.js, Postgres/Prisma, NextAuth, preview-deploy CI/CD; built via agentic Claude Code workflows.

sans_ai_mock

Multi-mode Claude Code agent for AI mock interviews; measured per-mock API cost.

sans_cube

Real-time Bluetooth Rubik's-cube solve analyzer — BLE, live 3D rendering, CFOP/Roux phase detection.

More work

[sans_learning](#) (7.8x model-cost delta), [cut_sh](#) (ffmpeg scripting), [sans_yt_summary](#) (prompt-injection hardening).